

DAY 2 — Prompt Engineering & Copilot Studio Development

Module 1: Prompt Engineering

1. What is Prompt Engineering?

Prompt Engineering is the process of designing effective instructions for AI models to generate:

- Accurate responses
 - Context-aware outputs
 - Business-aligned decisions
 - Structured enterprise responses
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Why Prompt Engineering is Important?

Good prompts help AI:

- Reduce hallucinations
 - Improve accuracy
 - Follow enterprise policies
 - Maintain professional tone
 - Produce reliable outputs
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Core Components of Prompt Engineering

Component	Purpose
System Prompt	Defines AI behavior
User Prompt	User request
Context	Supporting information
Memory	Previous interactions
Constraints	Rules & limitations

2. System Prompts

What is a System Prompt?

A system prompt defines:

- AI personality
 - Behavior
 - Tone
 - Restrictions
 - Responsibilities
-

Example System Prompt

You are an Enterprise HR Assistant.

Rules:

- Respond professionally
- Use company HR policies only
- Avoid guessing information
- Escalate legal or sensitive issues to HR team
- Keep responses concise and accurate

What is this?

Real-World Example

Without system prompt:

- AI may generate random responses

With system prompt:

- AI behaves like enterprise HR executive
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Demo — System Prompt Behavior

Objective

Observe AI behavior changes.

Demo Steps

Step 1

Open:

- [Microsoft Copilot Studio](#)
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Step 2

Create AI agent:

- HR Assistant
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Step 3

Add System Instructions

Paste professional HR instructions.

Step 4

Test Questions

Ask:

“Can you increase my salary?”

Observe:

- Professional responses
 - Controlled tone
 - Escalation handling
-

3. Context Engineering

What is Context Engineering?

Context engineering means:

- Providing supporting business data
 - Supplying documents
 - Maintaining conversation state
 - Passing workflow information
-

Example

Without context:

“What is leave policy?”

AI may hallucinate.

With context:

- AI reads HR document
- Generates grounded answer

Types of Context

Context Type	Example
Conversation Context	Previous chat
Business Context	HR policy
User Context	Employee role
Workflow Context	Approval stage

4. Prompt Chaining

What is Prompt Chaining?

Prompt chaining means:

- One AI output becomes next AI input

Example Workflow

Step 1:

Summarize employee complaint

↓

Step 2:

Generate manager escalation email

↓

Step 3:

Create Teams notification

Enterprise Use Cases

- Ticket processing
 - Invoice approvals
 - Incident management
 - Email automation
-

Demo — Prompt Chaining

Objective

Create multi-step AI workflow.

Demo Steps

Step 1

Open:

- [Microsoft Power Automate](#)
-

Step 2

Trigger:

- Email received
-

Step 3

AI Action 1:

- Summarize email
-

Step 4

AI Action 2:

- Classify urgency
-

Step 5

AI Action 3:

- Generate escalation email
-

Step 6

Send Teams notification

Learning Outcome

Students understand:

- Sequential AI workflows
 - AI orchestration
 - Automation pipelines
-

5. Role-Based Prompting

What is Role-Based Prompting?

AI behaves according to assigned role.

Examples

Role	Behavior
HR Manager	Employee support
Security Analyst	Risk analysis
Financial Advisor	Financial summaries
IT Support Engineer	Troubleshooting

Example Prompt

Act as a senior IT support engineer.

Responsibilities:

- Troubleshoot Windows login issues

- Guide users professionally
- Escalate security incidents immediately
- Provide step-by-step instructions

What is this?

Benefits

- Better specialization
 - Domain-focused responses
 - Professional outputs
-

6. AI Response Optimization

Optimization Techniques

Technique	Purpose
Clear instructions	Reduce ambiguity
Examples	Improve consistency
Constraints	Prevent hallucinations
Formatting rules	Structured output
Context grounding	Better accuracy

Example Optimized Prompt

Summarize the HR leave policy.

Requirements:

- Maximum 5 bullet points
- Use simple language
- Mention maternity leave separately
- Avoid technical jargon

What is this?

Module 2: Copilot Studio Deep Dive

1. Topics

What are Topics?

Topics are:

- Conversation flows
 - Intent-based workflows
 - Structured AI interactions
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Example Topics

Topic	Purpose
Leave Request	HR workflow
Password Reset	IT support
Payroll Query	Finance support

Topic Flow Example

User:

“Reset my password”

↓

Bot:

- Collect employee ID
 - Validate identity
 - Trigger reset workflow
-

2. Variables

What are Variables?

Variables store:

- User inputs

- Workflow values
- AI outputs
- Session data

Examples

Variable	Value
EmployeeName	Ajay
TicketID	INC1001
LeaveDays	5

Variable Use Cases

- Workflow tracking
- User personalization
- Multi-step conversations

3. Entities

What are Entities?

Entities extract structured information from user text.

Example

User says:

“Book leave on Friday”

Entity extracted:

- Date = Friday

Common Entities

Entity	Example
Date	Tomorrow
Email	user@company.com
Location	Bangalore
Number	5 days

4. Adaptive Cards

What are Adaptive Cards?

Adaptive Cards are:

- Interactive UI cards
 - Rich response components
 - Used in Teams & Copilot
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Features

- Buttons
 - Forms
 - Images
 - Approvals
 - Dropdowns
-

Example Use Cases

Use Case	Example
Leave Approval	Approve/Reject buttons
IT Ticket	Incident card
Employee Onboarding	Form collection

5. Generative Orchestration

What is Generative Orchestration?

AI dynamically:

- Chooses workflows
- Calls tools
- Uses context
- Plans actions

Without predefined hardcoded flows.

Example

User says:

“Schedule interview and notify panel.”

AI:

1. Reads calendar
 2. Finds slot
 3. Creates meeting
 4. Sends notifications
-

6. AI Actions

What are AI Actions?

Actions connect AI with:

- APIs
 - Workflows
 - Databases
 - Enterprise systems
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Examples

Action	Example
Create Ticket	ServiceNow

Action	Example
Send Email	Outlook
Create Meeting	Calendar
Update Record	Dataverse

Demo 1 — Build HR Support AI Agent

Objective

Create enterprise HR assistant.

Demo Steps

Step 1

Open:

- [Microsoft Copilot Studio](#)
-

Step 2

Create Agent:

- HR Support Assistant
-

Step 3

Configure:

- Greetings
 - Instructions
 - Professional tone
-

Step 4

Create Topics:

- Leave policy
- Payroll

- Benefits
-

Step 5

Add Generative AI:

- Enable AI responses
-

Step 6

Add Knowledge Source:

- HR PDFs
 - SharePoint
-

Step 7

Test Questions

Ask:

“What is maternity leave policy?”

Demo 2 — Create AI-Powered IT Helpdesk Assistant

Objective

Build enterprise IT support AI.

Demo Steps

Step 1

Create:

- IT Helpdesk Agent
-

Step 2

Add Topics:

- Password reset
- VPN issue

- Outlook problem
-

Step 3

Add Variables:

- Employee ID
 - Device Name
-

Step 4

Create Workflow:

- Ticket creation
-

Step 5

Integrate:

- Power Automate
-

Step 6

Test Scenario

Ask:

“My Outlook is not opening.”

Observe:

- AI troubleshooting
 - Ticket workflow
 - Escalation process
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Module 3: Knowledge Grounding

1. SharePoint Grounding

What is SharePoint Grounding?

AI retrieves enterprise knowledge from:

- SharePoint documents
 - Policies
 - Wikis
 - Knowledge bases
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Enterprise Benefits

- Grounded responses
 - Reduced hallucinations
 - Real-time enterprise knowledge
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2. Dataverse Integration

What is Dataverse?

Microsoft Dataverse is enterprise business data platform.

Use Cases

Scenario	Example
HR Records	Employee data
Tickets	IT incidents
Finance	Invoice tracking

3. Enterprise Document Grounding

AI uses:

- PDFs
- Word docs
- Policies
- SOP documents

As trusted knowledge sources.

4. RAG Architecture Basics

What is RAG?

RAG = Retrieval-Augmented Generation

AI:

1. Retrieves enterprise documents
 2. Sends context to GPT
 3. Generates grounded response
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RAG Workflow

User Query → Vector Search → Relevant Documents → GPT Response

RAG Benefits

- Enterprise grounding
 - Accurate responses
 - Reduced hallucination
 - Context-aware AI
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Module 4: AI Agent Design Best Practices

1. AI UX Design

Best Practices

- Clear responses
 - Professional tone
 - Guided interactions
 - Minimal confusion
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Good UX Example

Instead of:

“Invalid request”

User:

“I could not identify your employee ID. Please provide valid ID.”

2. Conversation Flow Design

Design Principles

Principle	Purpose
Short flows	Better experience
Guided prompts	Reduce confusion
Clarification handling	Better accuracy
Multi-turn support	Context continuity

3. Error Handling

Common Errors

Error	Handling
Missing input	Ask clarification
Invalid request	Provide guidance
API failure	Retry/escalate
Unknown query	Human handoff

Example

User:

“My payroll failed”

AI:

- Validate employee ID
 - Check payroll system
 - Escalate if needed
-

4. Human Handoff Scenarios

When AI Should Escalate

Scenario	Escalate To
Legal issue	HR Manager
Security threat	Security Team
Complex finance issue	Finance Team

Best Practice

AI should:

- Detect uncertainty
 - Transfer conversation smoothly
 - Preserve conversation context
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Case Study — AI Service Desk Automation Solution

Business Problem

Enterprise IT team faces:

- Thousands of tickets
 - Slow response times
 - Repetitive issues
 - Manual escalation
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AI Solution

Build:

- AI-powered service desk assistant

Using:

- Copilot Studio
- Azure OpenAI

- Power Automate
 - Dataverse
 - Graph API
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AI Capabilities

Capability	Example
Ticket creation	Auto-generate incidents
Troubleshooting	AI-guided fixes
Escalation	Human handoff
Knowledge retrieval	SOP documents
Notifications	Teams alerts

Enterprise Benefits

- Faster ticket resolution
- Reduced support workload
- 24/7 IT support
- Better employee experience